Homework 6: 09/24 Content

MATH 1271, Garrett Credi

09/24/2026

Problems listed as AC are found in the Active Calculus textbook at the end of the section listed.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed section.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Computing Derivative Functions

Choose 4 of the following problems, ones that you *did not* choose for the previous homework, to solve.

1. PAULS-ASSIGN.Derivatives.The Definition of the Derivative.1-24 (each of these count as one problem)

2 Type 2 - Graphically Interpreting Derivative Functions

Choose 3 of the following problems to solve.

1. AC 1.4.1
2. AC 1.4.2

3. AC 1.4.5
4. AC 1.4.10
5. AC 1.4.12

3 Type 3 - Determining Information from Derivative Functions

Choose 3 of the following problems to solve.

1. PAULS-ASSIGN.Derivatives.Interpretation of the Derivative.13,20,21 (each of these counts as one problem)
2. APEX.2.1.35
3. APEX 2.1.36

4 Challenge Problem

Compute the derivative functions for the following functions:

1. $f(x) = x^1$
2. $f(x) = x^2$
3. $f(x) = x^3$
4. $f(x) = x^4$

Do you notice a pattern with the derivative functions you find?

Use the above calculations to guess what the derivative function of *the power function*, $f(x) = x^n$ where $n > 0$ is a positive integer, is. Then, using the limit definition of the derivative, try and show that your guess is actually the correct formula for $f'(x)$. You will, in some way shape or form, need to expand the binomial power $(x + h)^n$ - think about how many terms you *actually need* and how you might go about determining the coefficients of those necessary terms.